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Autonomous University of Ciudad Juárez

Robotics Laboratory

Department of Electrical and Computer Engineering



semester project

Motors and controllers
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1. Introduction

This project presents the development of an underactuated biped based on five-part mechanisms bars at each end, with a mechanical design modeled in CAD and an assembly perfected through successive stages of adjustment and improvement of the prototype aimed at achieving Controlled cyclical movements and initial walking tests. An Orange is also used. Pi as the main computer and an integrated IMU to record orientation and accelerations during static posture, cycloid trajectories and walking.

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2. 5-bar linkage

Two five-bar mechanisms arranged in parallel were implemented, one at each end.
This is an underactuated mechanism with two degrees of freedom. The active joints
They are actuated by a pair of MG996R servomotors, while the joints
Passive motion was achieved using ball bearings. Two more servomotors were added per limb, in
the connection to the hip and the connection to the robot's foot.

2.1. CAD Design

Figure 1 shows one of the five-bar mechanisms in isolation.
corresponding to the upper part of the limb. On the other hand, Figure 2 shows
the five bars with a color code, described in Table 1, along with the lengths of
each expressed in millimeters (mm). Once the CAD design of the mechanism is finished
With 5 bars, the design of the entire bipedal robot, shown in Figure 3, was completed.

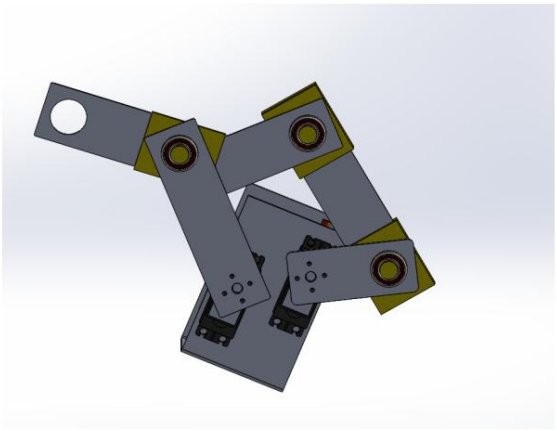


Figure 1: Design 1.

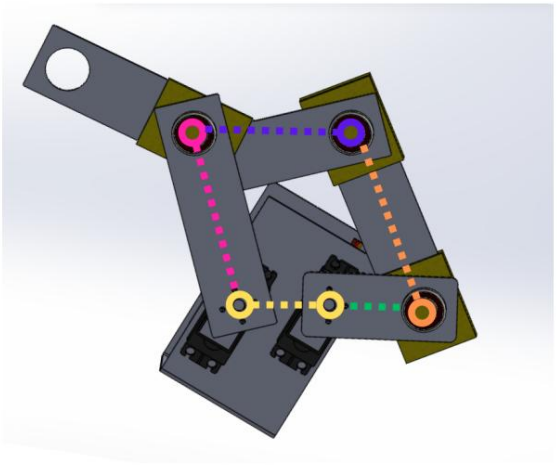


Figure 2: Definition of the 5 bars.

2.2. Kinematics of the mechanism

Once the five bars of the mechanism have been defined, the kinematic analysis is carried out with the
objective to determine the position, in millimeters, of the upper tip of the mechanism from
of the motor angles ÿ1 and ÿ4, as shown in Figure 4.

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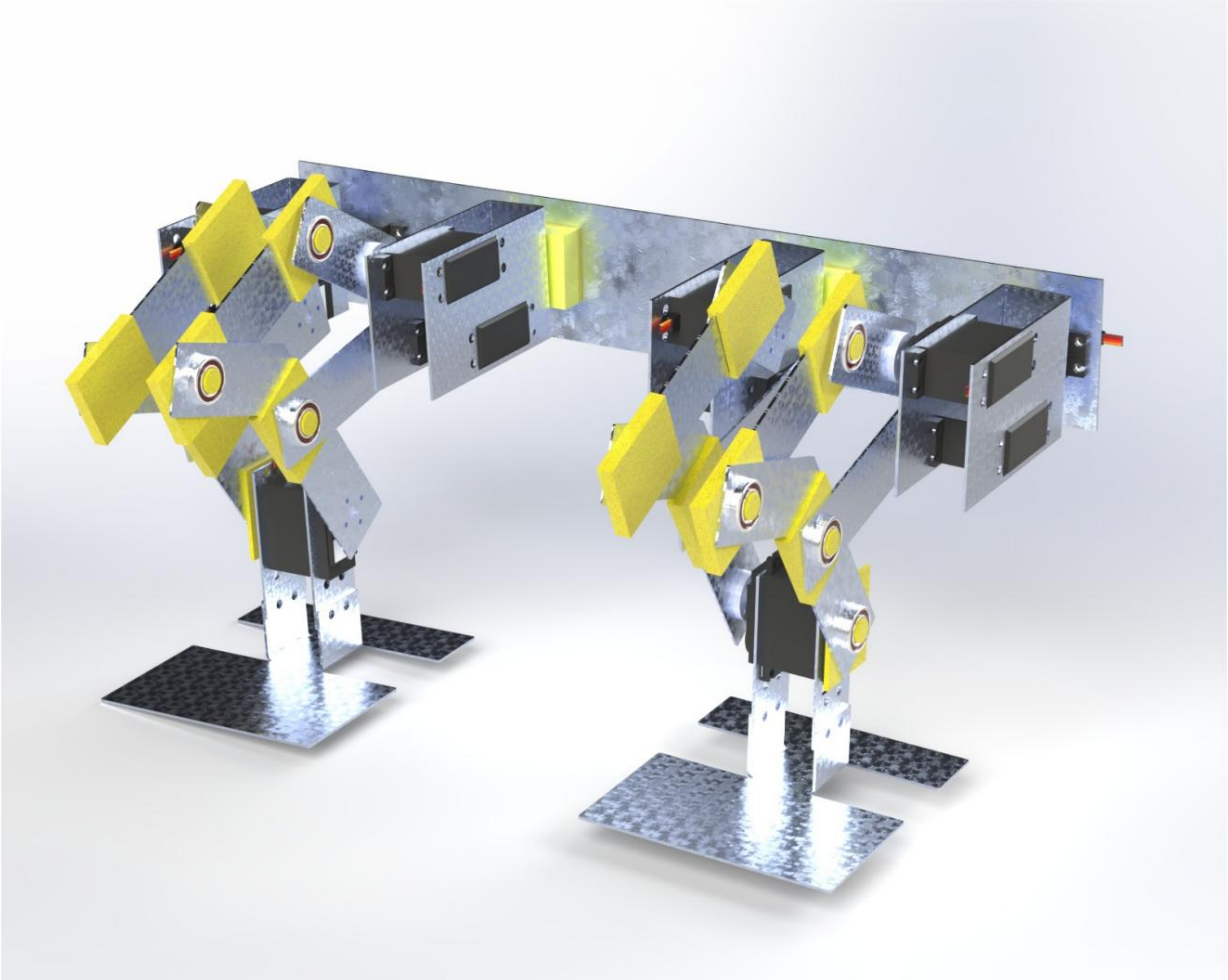


Figure 3: Design rendering.

Table 1: Table Title

Bar	Length	Color
L0	40.84 mm	Yellow
L1	43.5 mm	Green
L2	88.67 mm	Orange
L3	67.07 mm	Purple
L4	83 mm	Pink

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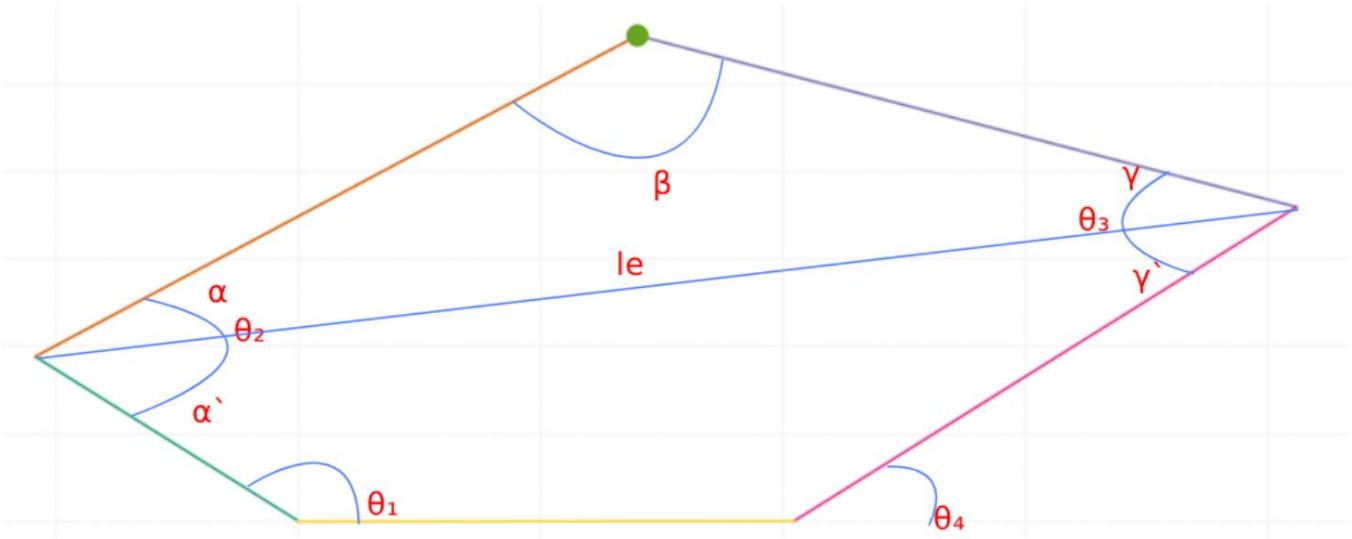


Figure 4: Visualization and nomenclature of the angles of the mechanism.

We begin by defining the position on the X and Y axes of the end of link L1 and L4 by the following expressions:

$$X1 = L1 \cos(ÿ1); \quad Y1 = L1 \sin(ÿ1);$$

$$X4 = L4 \cos(ÿ4); \quad Y4 = L4 \sin(ÿ4);$$

These positions are necessary to know the length of our "imaginary bar" Le, defined by:

$$Le = \sqrt{(X1 ÿ X4)^2 + (Y1 ÿ Y4)^2}$$

This allows us to see if the combination of angles ÿ1 and ÿ4 is possible without breaking our mechanism, because if the sum of L2 and L3 is less than our newly calculated Le, then The motors would be unable to reach the indicated angles without destroying the mechanism or themselves same.

Continuing with the analysis, we can use the law of sines and cosines and rearrange it to find

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the interior angles ÿ, ÿ and ÿ with the expressions:

$$ÿ = \arccos\left(\frac{Le^2 ÿL2^2 ÿL3^2}{ÿ2 \times L2 \times L3}\right)$$

$$ÿ = \arcsin\left(\frac{L3}{You} \times \sin(ÿ)\right)$$

$$ÿ = \arcsin\left(\frac{L2}{You} \times \sin(ÿ)\right)$$

To continue, we need to define another triangle by drawing an imaginary line, which
It goes from the top tip of L1 to the right tip of L0, as can be seen in the figure
5.



Figure 5: Visualization of L01 as the dotted line.

And using the law of cosines we can find the length of this imaginary line.

$$L01 = \sqrt{L0^{2} + L1^{2} - ÿ2L1L0\cos(ÿ1)}$$

with which we can calculate the local version of the angle ÿ using:

$$ÿ' = \arccos\left(\frac{L01^2 ÿL4^2 ÿLe^2}{ÿ2LeL4}\right)$$

which helps us to find the internal angle ÿ3 by adding ÿ with ÿ ' and we will also need
the complement of these angles. With which we can finally arrive at the positions
Cartesian coordinates of the end effector of the 5-bar linkage with:

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$$PX = L0 + L4 \times \cos(ÿ4) + L3 \times \cos(ÿ4 + ÿ3 \quad ')$$

$$Py = L0 + L4 \times \sin(ÿ4) + L3 \times \sin(ÿ4 + ÿ3 \quad ')$$

Once we know a condition that limits our mechanism, we can calculate all the possible locations for our mechanism and to have our workspace secure, shown in figure 6

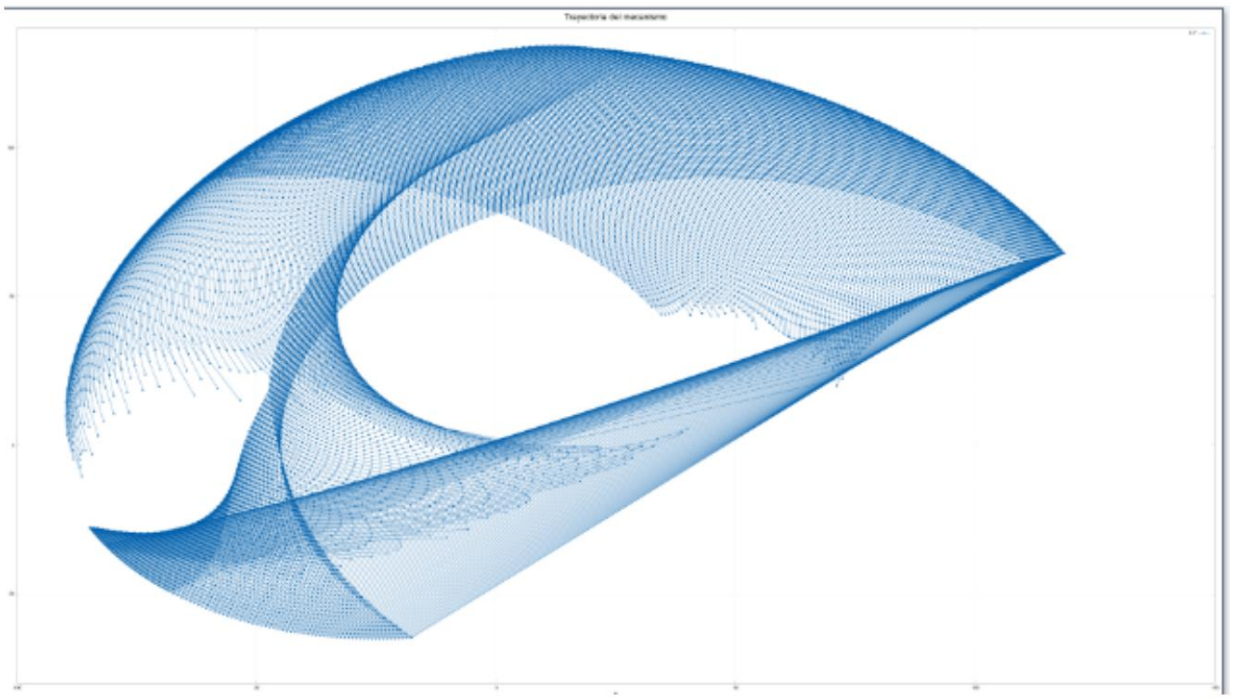


Figure 6: Workspace.

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3. Cycloid

The cycloid is the shape of motion that the five-bar linkage follows to simulate a hike.

In order for the mechanism to follow the desired movement, the following equations were applied to function of time (t):

$$ÿ1 = a+c×\sin(t) \quad (1)$$

$$ÿ4 = b+c×\sin t + 2 \quad \frac{ÿ}{—} \quad (2)$$

where:

a : Initial angle of ÿ1,

b: Initial angle of ÿ4,

c: Oscillation amplitude.

3.1. Servo Movement

Once the joint movements were defined, the commands were sent to the servomotors. using an ESP-WROOM-32 and an expansion card with power supplies independent for control and power.

3.1.1. **Master** - **Slave** Communication

The computer embedded in the biped sends movement signals via serial port using the computer's USB ports and the control and sensing card.
A telegram is sent specifying which servomotor will be moved to what angle and with That's fast.

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3.1.2. Angular velocity

For controlling the angular velocity of the servos, the initial choice was to use

Libraries were used, but these did not provide good control of multiple servos simultaneously. To solve this

This led to the development of an algorithm;

$$\ddot{y}t = \ddot{y}0 + \sum_{k=1}^{N} \ddot{y}\, \mathrm{sgn}(\ddot{y}i+N\, \ddot{y}\ddot{y}i(k\, \ddot{y}1))\ddot{y}\ddot{y}i$$

Where ÿÿi are the angular intervals taken by the servo, calculated by:

$$\ddot{y}\ddot{y}i = \frac{1}{\mathrm{veli}}$$

The time intervals were calculated using a microcontroller time counter,

instead of using interrupts, so that multiple servos can be controlled simultaneously.

All of this was done to create a constant and smooth speed in the servomotors, by sending them

to these angles by means of pulse width modulation, since it is the only data they can

receive the servos.[1]

3.2. Generated cycloid

Once we understand the kinematics of our mechanism and have control of the actuators

From this, the cycloid function was used and several iterations were performed in the

parameters of equations 1 and 2 until a response is obtained that allows the robot to move

maintaining its balance. The response of the end effector of the 5-bar linkage is shown

in figure 7

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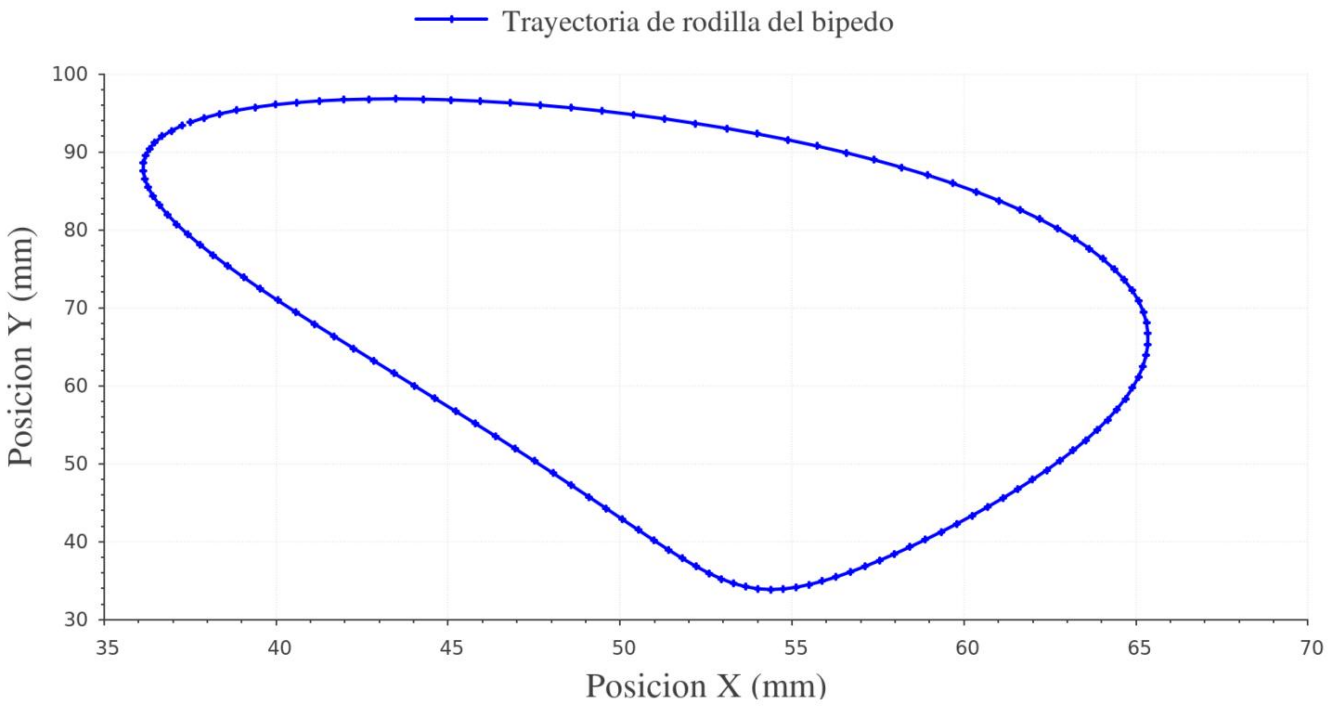


Figure 7: End effector of the 5-bar linkage.

4. Sensing

To prevent the Biped from falling, it needs to know that it is falling, so it is

I implement an IMU.

The selected IMU was the BNO80X, whose sensors are shown in Table 2

4.1. Data Fusion

The BNO08x is a System in Package (SiP) that includes a dedicated microcontroller and its

its own sensors. The secret to its performance lies in the fact that the microcontroller executes the fusion algorithm within the chip, without the need for intervention from the main processor of the user. [2]

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Table 2: IMU BNO08X

SENSOR	MEASURES	MAIN PROBLEM
Accelerometer	Gravity and linear acceleration.	It is sensitive to vibration noise.
Gyroscope	Angular velocity (rotation).	He/She suffers from drift.
Magnetometer: Measures the Earth's magnetic field (compass). It is very sensitive to interference from metals.		

4.1.1. Master - **Slave** Communication

Once our IMU is defined, it will communicate with the ESP-32 using the I2C protocol, and It will be constantly reading data from the IMU. Until the Orange Pi sends it a send a telegram requesting the IMU data, then the ESP-32 will send the latest data that has been stored in memory. This communication algorithm is visually exemplified in the Figure 8.

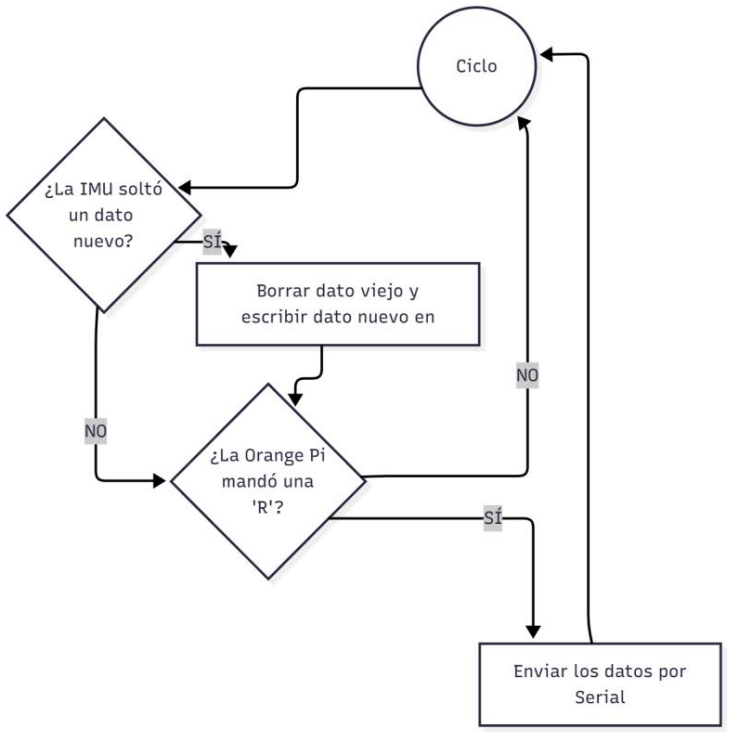


Figure 8: Slave ÿ Master communication sequence diagram.

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5. Control Strategy

The robot actively monitors its tilt in real time to make safety decisions, alternating between a Walking mode and a Recovery mode.

The control algorithm implemented in C++ executes the following logical loop in each iteration, also shown in Figure 9:

1. **Data Acquisition (Feedback):** The computer sends the data acquisition telegram data to the sensing module. The IMU subsystem responds by sending the angle of Current pitch.

2. **Stability Assessment:** The algorithm checks if the robot is inside a safe operating range, experimentally defined as ±20° of inclination in the Pitch axis.

3. **Decision Making:**

   - **Stable Case** (|Pitch| < 20°): If the pitch is safe, the system executes the next step of the cycloid trajectory. The coordinates (X,Y) are calculated, It solves the inverse kinematics and the servos move forward.

   - **Unstable Case (Fall Detected):** If the threshold is exceeded, the system is interrupted. temporarily stops the generation of the cycloid and activates a Safety Reflex:

     • If Pitch > 20° (Frontal Drop): The free limb moves rapidly forward to expand the support polygon.

     • If Pitch < ÿ20° (Backward Drop): The limb moves backward to counterbalance the fall.

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4. **Resumption of Movement:** The system maintains the safety position until the IMU readings return to the ±20° range . Once stabilized, the algorithm resumes the execution of the cycloid walk from the point where it was interrupted.

This approach allows for a predictable cyclical walk, delegating stability to the design. mechanical and passive filtering of disturbances.

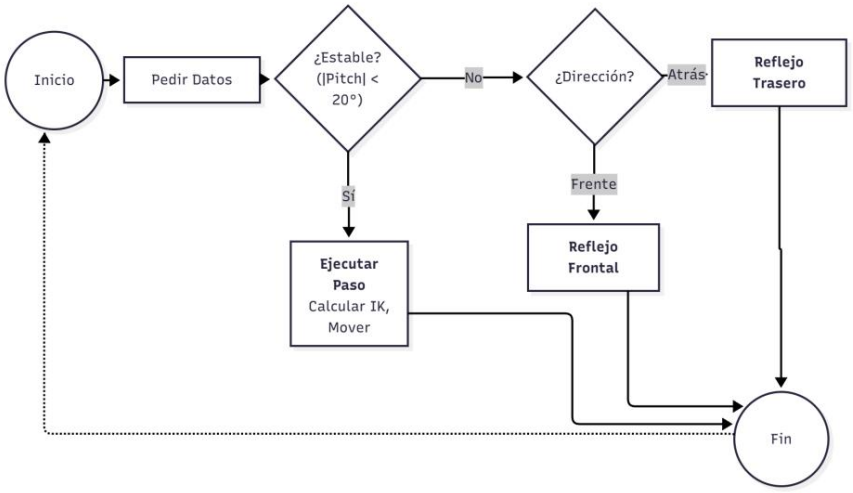


Figure 9: Walk sequence diagram

With this control architecture, even though it is somewhat "primitive", he managed to take a walk in the that the biped remained stable without needing support, thus managing to walk alone. [3].
While the walk was taking place, the IMU data was stored for later use.
graphed, which can be seen in figure 10

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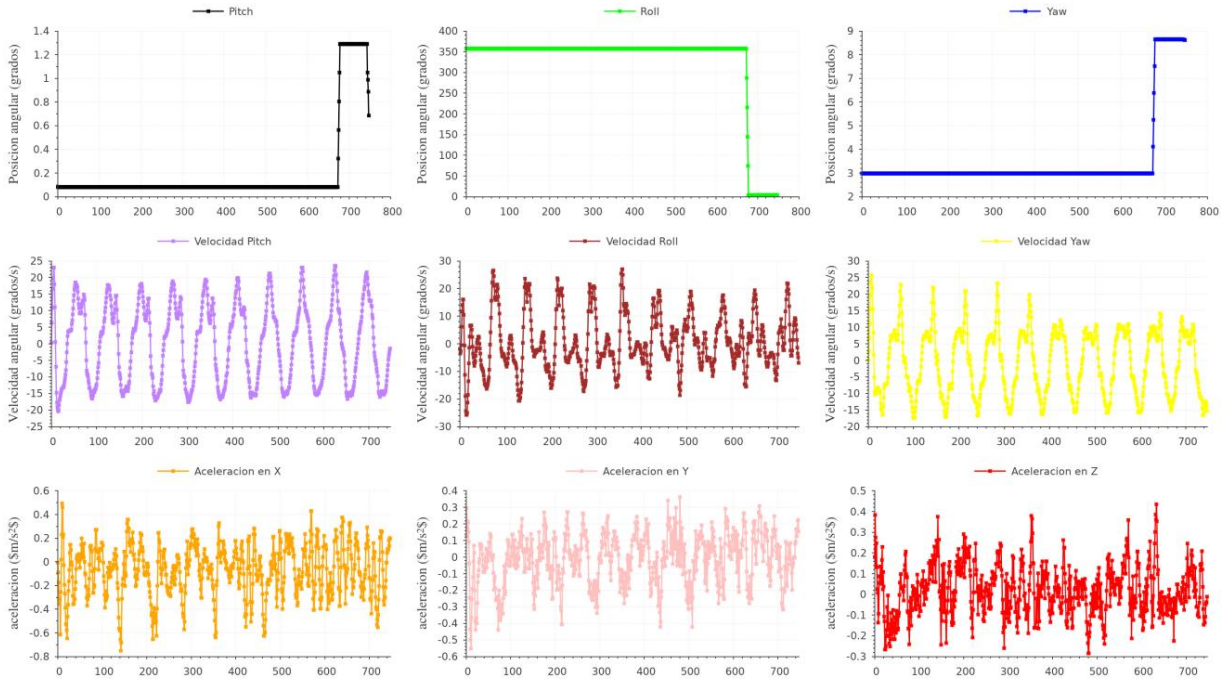


Figure 10: Walk sequence diagram

6. Conclusions

- Conclusion 1: The project to be carried out was the development of a bipedal walking robot, implementing a five-bar architecture for system kinematics, but
  We had problems with the structure of the biped due to its dimensions and weights.
  It was able to withstand it; the longest it took us to work on the biped was on the cycloid.
  that while we were looking for the right angles we had to be using trial and error to
  that he would walk the way we wanted, so that he would stay upright and
  could follow its path without tripping over its weight

- Conclusion 2: In summary, the project was a great experience working on making the biped since it requires solving both mechanical engineering problems (weight and structure) such as the precision in the implementation of mathematical functions (the cycloid) for to have a good walk.

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- Conclusion 3: Overall, the biped project achieved its main objective:
  generate a stable walk from a five-bar linkage controlled by
  a simple architecture, although there are still many improvements to be implemented.

  During development, several opportunities were identified that could optimize both the
  Performance, such as system efficiency, is a factor. For example, the capacity of
  neither the multiple serial ports available on the Orange Pi nor on the ESP32, which would have
  allowed to separate communication from sensing and control of servomotors, avoiding
  communication bottlenecks and potential blockages.

  Nor was the potential of the two cores of the ESP32 or the Orange Pi exploited, where
  One core could have been assigned exclusively to data acquisition and another to
  motion control, achieving truly parallel and smoother control.

  Another pending issue was the implementation of a Kalman filter, which would have allowed
  smooth the IMU data and generate a more accurate and controlled gait, instead of
  relying on a somewhat "brute force" strategy.

  Finally, I would have liked to perform a more in-depth dynamic analysis using the
  Lagrange's formalism, so that the biped could better take advantage of gravity and achieve a
  more natural and efficient movement.

  Even with these limitations, the project served as a solid foundation for understanding the
  integration between the mechanical, electronic and control parts of a bipedal robot, and leaves a
  a good starting point for more advanced versions in the future.

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